YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 7 OF 10 · 40 MINUTES

The ice ship mission 冰船计划

Fibres make a brittle material into a composite that is harder to break; a good plan lets us test this fairly.

- I can plan a fair test: question, variables, prediction with a reason, method and risks.
- I can make identical samples that change only one thing, and label them clearly.

composite material 复合材料 · reinforce 加固 · fibre 纤维 · brittle 脆的 · strength 强度 · sawdust 木屑 · mould 模具 · sample 样品 · hazard 危险源 · risk 风险

THE LESSON

Do Now	4	Folders back; Week 6 retrieval; Lead Scientist reads plan page 1; today's goals. Slides 1 to 2.
Hook	3	Pykrete story: Pyke's wartime ice ship; predict why sawdust makes ice stronger. Slides 3 to 4.
Learn	6	How fibres stop cracks; questions, tests; frame tour; model a risk row; Core A. Slides 5 to 10.
Do part 1 (plan)	13	Groups finish frame in roles; teacher approves samples on the board; Core B, Stretch. Slide 11.
Think and Reflect	4	Plan swap with checklist and one fix; teacher signs 'Plan approved'; exit ticket. Slide 12.
Do part 2 (make samples)	10	Safety slide; pairs make and label about 9 strips per group; 2 min tidy. Slides 13 to 17.

Hands-on: Pairs make, label about 9 fibre strips per group in foil troughs: ice (Plan A) or salt dough (Plan B).

Collect or assess: Approved planning frame (photographed); samples logged on class board; exit ticket: sample code, a control with reason.

SAFETY

- Both: untreated sawdust only, never MDF. Plan A: keep damp, water first, never tip
- Plan A: mop spills at once; only the teacher carries trays and uses the freezer
- Plan B: allergic students never touch dough; never eat it; knead in bag; wash hands

WATCH OUT

- Every group includes the control (plain ice or 0 g) and changes one thing; check the board first.
- Strip trial done before this lesson; labels in pencil, never cut labelling; trays flat and level.

Language: Frame: 'I predict ___ because the fibres ___.' Band 1: buddy, Core B first. Mould is the container, not fungus.

Next week: Week 8 test day: samples from freezer or shelf. Get books, string, bags, jugs, trays, towels, gloves, mop.

PREP FOR ONE CLASS

Plan A: before the lesson about 120 min (95 could be done by someone else); after about 27 min (some times estimated)

Plan B: before the lesson about 130 min (110 could be done by someone else); after about 35 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Check handover note for Plan A or B; send that plan's print job
  5 min · reprographics

THE WEEK BEFORE

- [ ] Buy fibres, foil, bags, paper cups, tape, kitchen roll; Plan B: flour, salt
  15 min · technician
- [ ] Plan A: book a freezer drawer to Week 8; clear, label, level it day before Plan A 10 min · teacher
- [ ] Plan B: check allergy list; give allergic students a non-dough role
  Plan B 5 min · teacher
- [ ] Read Week 6 drafts, list portions; day before, put up board, trial decision 15 min · teacher
- [ ] Fold 60 foil troughs on Week 5 block; cut 64 labels, 10/group; prep fibres 45 min · technician
- [ ] Laminate the A3 class project board (optional) optional
  5 min · technician

THE DAY BEFORE

- [ ] Plan A: pre-portion and label fibres (up to 36); sawdust damp, sealed
  Plan A 20 min · technician
- [ ] Plan B: pre-portion and label fibres (up to 36); clear a dry shelf
  Plan B 15 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Plan B: make 2 dough batches; weigh 54 balls of 18 g plus spares, bag each Plan B 20 min · technician
- [ ] Set out 6 kit trays and group folders (Plan A: mop ready)
  10 min · technician

AFTER THE LESSON

- [ ] Plan A: carry trays flat to the freezer; freeze at least 24 hours
  Plan A 5 min · teacher
- [ ] Plan B: trays flat to drying shelf; on day 3 peel foil, turn strips
  Plan B 13 min · teacher
- [ ] Store kit and spare fibre; photograph approved plans; read exit tickets 20 min · teacher

PRINT

- Worksheet, Plan A or B x24 (2 A4 sheets each; page 3 single-sided)
- Sample labels x2 A4 sheets (64 labels, single-sided)
- Class project board, Plan A or B x1 (A3, single-sided)
- Teacher copy: kit card for your plan x1 (A5, double-sided)

GET OR MAKE (NEW THIS LESSON)

- 25 ml cylinders or 20 ml syringes x12 (Plan A)
- Small mixing cups x18, water cups (200 ml) x6 (Plan A)
- Teaspoons x18
- Flat trays for 9 troughs x6 (Week 6 trays if flat)
- Untreated sawdust, never MDF (up to 30 g, per plans)
- Tissue, shredded paper, cotton wool (up to 12 g each)
- Kitchen foil x60 pieces, about 20 cm x 10 cm
- Small paper cups for fibre portions x36
- Small zip-lock bags x54 for dough balls (Plan B)
- Labelled bags for spare fibre (1 per fibre type)
- Paper towels x24 sheets
- Masking tape (6 rolls on requisition)
- Plain flour 500 g and salt 580 g (Plan B)
- Mixing bowl, cup measure, damp cloth (Plan B)
- Freezer drawer (Plan A) or dry shelf (Plan B)
- A3 laminating pouch x1 (optional)

Have ready from earlier weeks: Group folders x6, planning frames, 2 spare planning frames, Role card sets x6, Sentence frame card sets x6, Science Dictionaries x24, Class box; logbooks x24 (optional), Pencils x24, marker pen, Scale, template block; Plan A jug, mop